
Unit – 2 : Session – 7 : SLO – 1 | Multithreading Models — Fill in the Blanks

Multithreading

1. The user-level threads model manages threads without kernel intervention, resulting in faster context switching.
2. Kernel-level threads are threads managed directly by the operating system kernel.
3. The hybrid model combines user and kernel-level threading to gain benefits of both.
4. Multithreading improves concurrency by allowing multiple threads to run simultaneously or appear to run concurrently.
5. One advantage of multithreading is better resource sharing of resources within a process.
6. Frequent context switching can cause performance degradation due to CPU time spent switching between threads.
7. One disadvantage of kernel-level threads is increased overhead due to kernel involvement.
8. User-level threads have better portability across different operating systems since they do not depend on kernel support.
9. A major problem in multithreaded programs is deadlock, where two or more threads wait indefinitely for resources.
10. Parallelism allows a program to run efficiently on multiple CPU cores.